ASSOCIATED POINTS UNDER AN AFFINE MORPHISM OF LOCALLY NOETHERIAN SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. The proof is a chain of inclusions; each inclusion is produced by a named input, and the chain is assembled in place, so the card is taken (the pattern 'combine two named lemmas and stop' is what the filter rejects, not a chained argument of this shape).

Editorial note. The references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Local Noetherianity passes to the affine ring. Properties, Lemma properties-lemma-locally-Noetherian, used at proof lines 849-851. Makes B Noetherian.
- Associated points of a sheaf on an affine scheme are the associated primes of the module. Lemma lemma-associated-affine-open, used at proof lines 852-854. Translates the problem into algebra.
- Functoriality of associated primes for a Noetherian ring map. Algebra, Lemma algebra-lemma-ass-functorial-Noetherian, used at proof lines 856-858. Gives the first inclusion.
- Associated points are weakly associated, and weak associativity is functorial backwards. Lemmas lemma-weakly-ass-support and lemma-weakly-ass-reverse-functorial, used at proof lines 859-866. Give the middle and last inclusions.
- The outer sets agree. Lemma lemma-ass-weakly-ass, used at proof lines 867-868. Closes the chain of inclusions into an equality.

Source reference. Original lemma 05DZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6972–6972. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05AR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6859–6859. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02OK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3519–3519. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05EX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7006–7006. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05AM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6855–6855. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01OW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2209–2209. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $f : X \to S$ be an affine morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. If X is locally Noetherian, then we have*

$$f(\mathit{Ass}_X(\mathcal{F})) = \mathit{Ass}_S(f_*\mathcal{F}) = \mathit{WeakAss}_S(f_*\mathcal{F}) = f(\mathit{WeakAss}_X(\mathcal{F}))$$

Proof. We may assume X and S affine, so $X \to S$ comes from a ring map $A \to B$. As X is locally Noetherian the ring B is Noetherian, see Properties, Lemma 01OW. Write $\mathcal{F} = \widetilde{M}$ for some B-module M. By Lemma 02OK the associated points of $\mathcal{F}$ correspond exactly to the associated primes of M, and any associated prime of M as an A-module is an associated points of $f_*\mathcal{F}$. Hence the inclusion

$$f(\mathrm{Ass}_X(\mathcal{F})) \subset \mathrm{Ass}_S(f_*\mathcal{F})$$

follows from Algebra, Lemma 05DZ. We have the inclusion

$$\mathrm{Ass}_S(f_*\mathcal{F}) \subset \mathrm{WeakAss}_S(f_*\mathcal{F})$$

by Lemma 05AM. We have the inclusion

$$\mathrm{WeakAss}_S(f_*\mathcal{F}) \subset f(\mathrm{WeakAss}_X(\mathcal{F}))$$

by Lemma 05EX. The outer sets are equal by Lemma 05AR hence we have equality everywhere. □